

Role of DNA Damage in Melanogenesis

UVB irradiation is directly absorbed by cellular DNA, leading to the formation of DNA lesions, primarily **cyclobutane pyrimidine dimers** and **pyrimidine (6-4) pyrimidone photoproducts**. This DNA damage activates repair systems, at least in part through the tumor suppressor **p53 protein**.

In addition to DNA damage, UV irradiation affects plasma membrane lipids, resulting in the release of **diacylglycerol (DAG)**, which activates **PKC- β** to stimulate melanogenesis via **tyrosinase activation**.

An important link between DNA damage and melanogenesis is supported by several observations: - The action spectrum for tanning closely matches that for thymine dimer formation. - UV-induced melanogenesis is enhanced in pigment cells treated with **T4 endonuclease V**, an enzyme that specifically promotes repair of UV-induced DNA damage. - Agents that cause DNA damage—without other cellular effects—also stimulate melanogenesis, indicating that DNA damage alone can trigger this response.

The **p53 protein**, known as the "Guardian of the Genome," plays a central role in this process. Upon activation by DNA damage, p53 upregulates **tyrosinase mRNA and protein levels**, thereby enhancing melanogenesis. This suggests that **tanning is part of a p53-mediated adaptive response to DNA damage**, serving to protect the skin from future UV exposure.

Indirect Effects of UV Irradiation on Melanocytes

UV irradiation induces keratinocytes to release various paracrine factors that modulate melanocyte function. These factors can act individually or synergistically to influence melanocyte proliferation, dendricity, melanogenesis, melanosome transfer, and survival.

Table 70-2: Ultraviolet Irradiation-Induced Keratinocyte-Derived Paracrine Cytokines

KERATINOCYTE-DERIVED FACTOR	PROLIFERATION	DENDRICITY	MELANOGENESIS	MELANOSOMAL TRANSFER	SURVIVAL
Basic fibroblast growth factor	↑↑				
Endothelin-1	↑	↑	↑		
Interleukin-1 α /1 β	↓	↑	↓		
Adrenocorticotrophic hormone (ACTH)	↑		↑		↑
α -Melanocyte-stimulating hormone (α -MSH)	↑	↑	↑		↑
Prostaglandin E ₂ / Prostaglandin F ₂ α	↑	↑	↑	↑	
Granulocyte-macrophage colony-stimulating factor	↑		↑		
Nitric oxide			↑		
Tumor necrosis factor- α (TNF- α)			↓		
Nerve growth factor (NGF)		↑			↑

Notably, UV irradiation also increases levels of **TNF- α** and **IL-1**, cytokines that inhibit melanogenesis. This indicates a balanced epidermal response involving both stimulatory and inhibitory signals, ultimately resulting in **net increased melanogenesis and melanocyte proliferation** following UV exposure.

Melanocyte Aging and Photoaging

Epidermal melanocyte aging is influenced by both **genetic** and **environmental** factors. Chronic UV exposure (photoaging) contributes to alterations in melanocyte number, distribution, and function, impacting skin pigmentation and increasing susceptibility to pigmentary disorders.